

MIE 402 - ME Lab II

Fall 2026

Instructor	Prof. Ehsan Roohi roohie@umass.edu Office: Gunness Laboratory room 1
Graduate Teaching Assistants	Yuyuan Zhang; Nazila Emamdoost (emails listed with lab sections below)
Theory Meeting	Mon. 12:20-1:10 PM Hasbrouck Laboratory room 134
Friday Office Hours	Fri. 12:20-1:10 PM Gunness Laboratory room 1

Course information

This course integrates lectures, pre-lab preparation, hands-on experiments, data analysis, and technical reporting. The Fall 2026 offering covers Laboratories 1 through 6 only.

References and software

No required textbook.

Figliola & Beasley, Theory and Design for Mechanical Measurements, 6th ed., Wiley, 2015.

Beckwith et al., Mechanical Measurements, 6th ed., Pearson, 2006.

Wheeler & Ganji, Introduction to Engineering Experimentation, 3rd ed., Pearson, 2009.

MATLAB and Simulink are required for selected pre-labs, analysis, and reports.

Grading

Component	Details	Weight
Pre-lab assignments	Six pre-labs, equally weighted	20%
Laboratory reports	Lab 1 report	10%
	Combined Labs 2 and 3 report	25%
	Combined Labs 4 and 5 report	25%
	Lab 6 report	15%
Participation	Lecture and laboratory participation	5%
Total		100%

Rubric for grading the reports

Complete sections (Abstract, Introduction (theory + experiment), Method, Results, Comparison, Discussion, Conclusions): 10%

Overall aesthetic: 10%

Clear figures: 10%

Abstract: 10%

Data analysis: 10%

Introducing the theory: 10%

Introducing the experiment: 10%

Comparison with theory: 10%

Discussion section: 10%

Meaningful results: 10%

Course policies

Each pre-lab assignment is due on Canvas before the student's own laboratory section begins. Solutions may be posted after the final section for that experiment; late pre-labs will not be accepted.

Reports are due on Canvas at 11:59 PM on the dates listed in the tentative course schedule. A report grade will be reduced by 25% for each day or partial day it is late.

Follow the laboratory report template and the stated page limit for each submission.

Do not use screenshots of the MATLAB (Python) figure window in pre-labs or reports. Export figures in a high-resolution format such as PNG, EPS, or PDF. Group related data in a single figure when comparison improves the analysis.

Students may discuss approaches to pre-labs, but copying solutions or sharing MATLAB (Python) or Simulink source files is prohibited. All submitted work must comply with the UMass Amherst Academic Integrity Policy.

Make-up laboratories are possible only when an instructor/GTA and appropriate space and equipment are available. Because equipment changes between experiments, a make-up may not be possible after the course moves to the next laboratory topic.

Notify the instructor during the first two weeks about university-approved schedule conflicts. Disability-related accommodation requests should be initiated with Disability Services as early as possible; accommodations are not retroactive.

Any pre-lab or report may include a brief oral verification. An inability to explain submitted work may result in loss of credit for that submission.

Ask questions of general interest during lecture and individual questions during office hours. Use email when neither option is practical. Clearly show work in every submission to receive partial credit.

Student support and university statements

Inclusivity, names, and pronouns

The diversity of participants in this course is a valuable source of ideas, problem-solving strategies, and engineering creativity. Everyone has the right to be addressed by the name and pronouns they use. If any aspect of the course prevents you from learning or makes you feel excluded, please contact the instructor so that we can develop strategies that meet both your needs and the requirements of the course. Concerns may also be shared with the College of Engineering or through its classroom experience and climate reporting channels.

Health and wellbeing

Your health and wellbeing are important to your success. If personal circumstances affect your performance, please contact the instructor as early as possible. Additional support is available from your academic advisor, the College of Engineering Office of Student Affairs, the Office of Diversity, Equity, and Inclusion, and campus support services.

Academic integrity

UMass Amherst is committed to academic integrity. Academic dishonesty includes cheating, fabrication, plagiarism, lying, and facilitating dishonesty through analog or digital means. Students are responsible for completing all work with integrity and in accordance with university policy. Suspected violations may be referred through the university academic-integrity process.

Policy: <https://www.umass.edu/senate/book/academic-regulations-academic-integrity-policy>

Disability accommodations

UMass Amherst is committed to equal educational opportunity. Students with a documented physical, psychological, or learning disability may be eligible for reasonable academic accommodations. Please notify the instructor within the first two weeks and contact Disability Services to make appropriate arrangements.

Disability Services: <https://www.umass.edu/disability/>

Title IX

UMass Amherst is committed to a safe learning environment free from discrimination, sexual assault, sexual harassment, domestic violence, dating violence, stalking, and retaliation. Title IX also protects against discrimination related to pregnancy and related conditions. Support is available without making a formal report.

For Title IX reporting purposes, the instructor is a responsible employee and must share disclosures of covered conduct with the Title IX Coordinator. Students decide whether and how to engage with that office, and retaliation for participating in a Title IX process is prohibited.

Resources: <https://www.umass.edu/titleix/resources> | Confidential SASA Hotline: 413-545-0800

Laboratory sequence

Lab	Experiment	Primary focus
1	Dynamic data sampling and frequency analysis	Sampling, aliasing, and spectral interpretation
2	Single pendulum	Dynamic response and model-data comparison
3	Double pendulum	Coupled motion and nonlinear behavior
4	1-DOF system - free response	Natural frequency and damping
5	1-DOF system - forced response	Frequency response and resonance
6	Radiometer	Radiometer movement measurement

Laboratory sections and GTA assignments

Section	Class no.	Meeting time	GTA	Location
01LL	15059	Tue. 2:30-5:15 PM	Yuyuan Zhang yuyuanzhang@umass.edu	Gunness Lab 5
01LM	15060	Wed. 9:05-11:50 AM	Nazila Emamdoost nemamdoost@umass.edu	Gunness Lab 5
01LN	15061	Wed. 1:25-4:10 PM	Nazila Emamdoost nemamdoost@umass.edu	Gunness Lab 5
01LQ	15066	Thu. 2:30-5:15 PM	Yuyuan Zhang yuyuanzhang@umass.edu	Gunness Lab 5
01LR	15075	Fri. 9:05-11:50 AM	Nazila Emamdoost nemamdoost@umass.edu	Gunness Lab 5
01LS	15100	Fri. 1:25-4:10 PM	Yuyuan Zhang yuyuanzhang@umass.edu	Gunness Lab 5

Tentative course schedule

Dates follow the UMass Amherst Fall 2026 calendar. Theory meets Mondays, Fridays are office hours, and there is no regular Wednesday theory meeting. All sections complete the same experiment during each listed laboratory week; dates remain tentative.

Wk	Dates	Pre-lab / report due	Monday theory / Friday office hours	Laboratory section
1	Sep. 8-11	-	Wed: one-time course orientation Fri: office hours	No laboratory
2	Sep. 14-18	-	Mon: measurement fundamentals Fri: office hours	No laboratory
3	Sep. 21-25	Pre-lab 1: before section	Mon: Lab 1 theory Fri: office hours	Lab 1 - all sections
4	Sep. 28-Oct. 2	Pre-lab 2: before section	Mon: Lab 2 theory Fri: office hours	Lab 2 - all sections
5	Oct. 5-9	Lab 1 report: Mon Oct. 5	Mon: No class Fri: office hours	No laboratory
6	Oct. 12-16	-	Mon: Indigenous Peoples' Day - no class Fri: office hours	No laboratory
7	Oct. 19-23	Pre-lab 3: before section	Mon: Lab 3 theory Fri: office hours	Lab 3 - all sections
8	Oct. 26-30	Pre-lab 4: before section	Mon: Lab 4 theory Fri: office hours	Lab 4 - all sections
9	Nov. 2-6	Labs 2 and 3 report: Mon Nov. 2	Mon: No class Fri: office hours Tue: Election Day - no classes	No laboratory

Wk	Dates	Pre-lab / report due	Monday theory / Friday office hours	Laboratory section
10	Nov. 9-13	-	Mon: No class Fri: office hours Wed: Veterans Day - no class	No laboratory
11	Nov. 16-20	Pre-lab 5: before section	Mon: Lab 5 theory Fri: office hours	Lab 5 - all sections
12	Nov. 23-27	Labs 4 and 5 report: Mon Dec. 7	Mon: No class Thanksgiving recess after Tue Nov. 24	No laboratory
13	Nov. 30-Dec. 4	Pre-lab 6: before section	Mon: Lab 6 theory Fri: office hours	Lab 6 - all sections
14	Dec. 7-11	-	Mon: No class Fri: office hours	No laboratory
15	Dec. 14-15	Lab 6 report: Mon Dec. 14	Mon: final office hour Classes end Tue Dec. 15	No laboratory

Important scheduling note. Pre-labs are due before each student's section. Section-specific changes or make-ups will be announced on Canvas.

Official academic calendar: UMass Amherst Registrar